(a) Aligned anchor positions

nl_base 11 tokens, prose

Ana has **12** apples ,

buys **5** more . Total **?**

(q_1 = 12; q_2 = 5; FINAL = ?)

⇕ same three anchors (q_1, q_2, FINAL), matched by meaning, not by token index

formal 21 tokens, bindings + prose

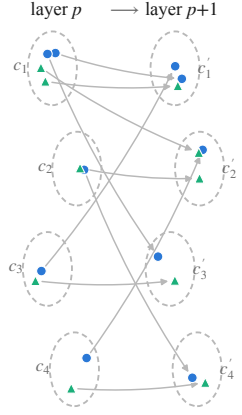
GIVEN n = **12** ; m = **5**

WHERE Ana has n apples , buys m more

FIND answer = **?**

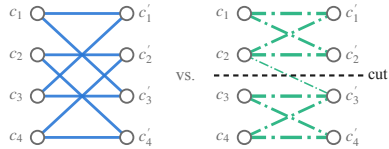
Anchor = each quantity-introduction site, plus the final token. Only anchor states enter the analysis, so length and surface form do not enter the graph.

(b) Clustering



$k = 4$ chosen ONCE per layer on nl_base by eigengap, then reused for every condition at that layer

(c) Transition graph and metrics



well-mixed, no good cut: λ_2 higher, entropy higher

a good cut exists: λ_2 LOW, entropy lower

λ_2 = algebraic connectivity of the pooled graph. LOW when a good cut exists (Cheeger): fewer, better-separated regions visited in sequence. H3 predicts formal − nl_base < 0. Path entropy = spread over transition paths, in $[0, \log k]$.

Fork layer = first layer at which the two conditions' cluster sets are DISJOINT (nl_base vs. nl_break)

	$\ell=0$	$\ell=1$	$\ell=2$	$\ell=3$	$\ell=4$	$\ell=5$	$\ell=6$
nl_base	{1,3}	{1,3}	{2,3}	{2,4}	{2,4}	{1,2}	{1,2}
nl_break	{1,3}	{1,2}	{2,4}	{1,3}	{1,3}	{3,4}	{3,4}

fork layer = 3: clusters shared before (shaded), none after. A pair that never forks is right-censored, not dropped.

Illustrative schematic with hand-chosen toy values ($k = 4$, one problem, a handful of states). Not data; results are in Figs. 1–3.